BMU Firmware V1012 — Alarm System User Guide

Product: ENKO BMU-52S
Firmware: V1012
Document scope: Alarm set parameters, alarm types, and operating guidelines for users.

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1. Overview

The Battery Management Unit (BMU) continuously monitors battery voltage, current, temperature, and auxiliary sensor readings. When a measured value crosses a configured threshold and remains there for the required confirmation delay, the corresponding alarm becomes active.

Startup hold: All alarm evaluation is suspended for **10 seconds** after system power-up to allow measurements to stabilize.

> **Parameter-gated evaluation:** Every alarm limit is gated. If the PROT or WARN parameter equals its own minimum value (i.e., the parameter is unconfigured), the alarm is held inactive regardless of the measured value. Setting the limit parameter to its minimum effectively disables the alarm without firmware changes.

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2. Alarm Set Structure

The BMU continuously monitors numerous measurements. For each monitored measurement that requires an alarm, the BMU uses an **alarm set**: a group of 7 parameters that together define exactly how the BMU detects, confirms, and clears a fault condition. Every alarm set contains the same 7 parameters, regardless of what is being measured.

#	Parameter	Short Name	Role
1	Protection	**PROT**	The critical limit — crossing this triggers a protective action
2	Protection Threshold	**PROT TH**	How far the value must recover before the protection alarm clears
3	Confirmation Delay	**DELAY**	How long (ms) the fault condition must persist before any alarm activates

#	Parameter	Short Name	Role
4	Trigger Counter	**NOT**	How many times protection can self-clear before requiring a manual reset
5	Counter Decay	**DECDLY**	How many hours of fault-free operation cause the trigger counter to drop by 1
6	Warning	**WARN**	The advisory limit **—** warns the operator before the protection level is reached
7	Warning Threshold	**WARN TH**	How far the value must recover before the warning alarm clears

The two alarm parameters (PROT and WARN) define *where* the alarm activates. The two threshold parameters (PROT TH and WARN TH) define *where* it deactivates. The delay (DELAY) controls *how quickly* it activates. The counter (NOT) and its decay (DECDLY) control what happens when the same fault repeats over time.

2.1 Protection (Prot.)

The protection is the hard limit for a measurement. When the measured value crosses this boundary, the BMU takes immediate protective action. The protection should be set at a value that, if sustained, could damage the battery or associated equipment. The system does not wait for operator intervention: it acts on its own as soon as the condition is confirmed.

Protection alarms can be disabled **(there are some exceptions)**. The alarm is disabled when the related parameter is set to its minimum value. When the alarm is disabled, digital outputs are not triggered.

> **Example:** A cell over voltage protection set to 4200 mV means the BMU will act the moment any cell reaches that voltage (after the confirmation delay).

2.2 Protection Threshold (Prot. Th.)

Once a protection alarm has activated **(not set its min value)**, the measured value must recover past the protection *by at least this amount* before the alarm is allowed to clear. This recovery margin is called the threshold value.

Without threshold, a value hovering right at the protection limit would cause the alarm to switch on and off rapidly. Threshold ensures a clean, stable transition: once an alarm activates, it stays active until the situation has clearly and fully resolved.

Setting the threshold to **0** is allowed, but it removes the recovery margin entirely: the alarm clears the instant the measured value drops back below the protection limit, with no buffer. This re-introduces the on/off chatter the threshold is meant to prevent, so a non-zero value is recommended for any measurement that can hover near its limit.

In some exceptional cases, the threshold feature is not used at all, regardless of its value. For example, sensor disconnection errors are not evaluated based on limit values, even if the threshold feature is set *(see §4.2)*.

> **Example:** Protection = 4200 mV, Protection threshold = 100 mV.
> Alarm activates at **4200 mV** → clears only when voltage drops below **4100 mV**.

2.3 Protection Delay (Prot. Delay - Confirmation)

The confirmation delay specifies how many milliseconds or seconds **(some of the alarms take the delay parameter's unit as seconds)** the fault condition must persist before any alarm in the set is confirmed and raised. If the condition resolves on its own before the delay expires, no alarm is recorded and no protective action is taken.

This parameter is shared: both the PROT and WARN alarms within the same alarm set use the same delay value **—** there is no separate WARN-only delay.

Unlike NOT, the delay cannot be disabled by setting it to 0. Every delay parameter carries a firmware-enforced minimum (100 ms for most alarms), so the shortest possible confirmation window is that minimum, not zero.

Short spikes in a measurement are suppressed by the delay. Only conditions that are sustained long enough to be genuine faults will trigger an alarm.

2.4 Number of Trial (NOT)

NOT applies only to the PROT side of the alarm set. The WARN alarm in the same set has no trigger counter of its own and always self-clears once its condition resolves, regardless of how many times it has fired.

The NOT parameter limits how many times a protection alarm can trigger and self-clear before operator intervention is required. Each time the protection alarm activates, an internal counter increments by one. When the counter reaches the configured limit, the alarm will no longer auto-clear. It must be manually acknowledged and reset by an operator.

Setting	Behavior
0	No limit **—** NOT disable
N	After N triggers, manual operator reset is required

An alarm that repeatedly triggers and self-clears often signals an intermittent fault that is getting worse. The trigger counter ensures that after a configurable number of recurrences, the fault is brought to the operator's attention rather than silently repeating in the background.

2.5 Decrease Delay (DECDLY)

Like NOT, DECDLY applies only to the PROT side of the alarm set — there is no equivalent decay counter for WARN, since WARN has no trigger counter to decay in the first place.

When a protection alarm has accumulated trigger counts **(via NOT)** but the fault has not occurred recently, the system can automatically reduce the counter over time. The DECDLY parameter sets how many consecutive hours of fault-free operation are required for the counter to drop by one step.

The DECDLY feature only has an effect when the NOT parameter is set greater than 0; if NOT is 0 (disabled), there is no counter for DECDLY to decay.

Setting	Behavior
0	The counter never decrements automatically
N	The counter decreases by 1 for every N hours the alarm remains inactive

DECDLY allows the system to gradually "forgive" infrequent faults. If a fault occurred once several days ago and has not repeated, the counter slowly returns toward zero over time.

2.6 Warning (Warn.)

The warning is set at a less extreme level than the protection, and is crossed first as a measurement moves toward a fault condition. When the warning limit is reached, the BMU notifies the operator **(through the SCADA)** but does **not** take any protective action.

The WARN alarm shares its alarm set's DELAY with PROT (section 2.3), but it has no NOT or DECDLY of its own **—** it always self-clears once the condition resolves and can never require a manual reset.

The warning gives the operator time to respond (reduce load, inspect the battery, investigate the cause) before the situation worsens and the protection threshold is reached.

For every monitored measurement, the warning threshold should always be set closer to the normal operating range than the protection threshold, so the warning reliably appears before protection is needed.

Warning alarms can be disabled **(there are some exceptions)**. The alarm is disabled when the related parameter is set to its minimum value. When the alarm is disabled, digital outputs are not triggered.

2.7 Warning Threshold (Warn. Th.)

Identical in concept to the protection threshold (Prot. Th.), but applied to the warning alarm. Once a warning has been triggered, the measured value must recover past the warning limit by at least this amount before the warning clears.

This prevents the warning indicator from flickering when a value is oscillating right at the limit boundary. As with the protection threshold, setting this value to **0** removes the recovery margin: the warning clears the instant the value drops back below the limit.

> **Example:** Warning threshold = 4100 mV, warning hysteresis = 80 mV.
> Warning activates at **4100 mV** → clears only when voltage drops below **4020 mV**.

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3. Alarm Reference Table

All 63 alarm ID slots (0–62) sorted by ID number. **Column guide:**

- **Type —** PROT = protection (takes immediate action); WARN = warning (notifies only, no action)
- **NOT —** ✓ means a trigger-count limit can be configured; once reached the alarm requires manual reset. Only available on PROT alarms
- **DECDLY —** ✓ means the trigger count can automatically decrease over time during fault-free operation. Only meaningful when NOT is also ✓

- Rows marked *(see §4.x)* use their PROT/WARN/TH fields differently than described in section 2 — see Non-Standard Alarm Logic

PBUSB = Positive Busbar , NBUSB = Negative Busbar , PMSD = Positive MSD Fuse , NMSD = Negative MSD Fuse , RB = Resistor Board , COM = Communication

> **Naming convention:** Alarm Name values match the `alarmComment` strings in the BMU parameter file (`paramBMU-52S.json` → `alarms[]`) so the two stay traceable to each other.

ID	Alarm Name	Type	Trigger Condition (Default)	NOT	DECDLY	Active During	V1012
0	No Fault	—	No errors found in the system	—	—	Always	✅
4	Pack Over Voltage Protection	PROT	Total pack voltage ≥ **192.5 V**	✓	✓	Always	✅
5	Pack Over Voltage Warning	WARN	Total pack voltage ≥ **190 V**	—	—	Always	✅
6	Cell Over Voltage Protection	PROT	Any single cell voltage ≥ **3700 mV**	✓	✓	Always	✅
7	Cell Over Voltage Warning	WARN	Any single cell voltage ≥ **3650 mV**	—	—	Always	✅
10	Pack Under Voltage Protection	PROT	Total pack voltage ≤ **137.5 V**	✓	✓	Always	✅
11	Pack Under Voltage Warning	WARN	Total pack voltage ≤ **140 V**	—	—	Always	✅
12	Cell Under Voltage Protection	PROT	Any single cell voltage ≤ **2650 mV**	✓	✓	Always	✅
13	Cell Under Voltage Warning	WARN	Any single cell voltage ≤ **2700 mV**	—	—	Always	✅
14	Charge Over Current Protection	PROT	Charge current ≥ **200 A**	✓	✓	Charging only	✅
15	Charge Over Current Warning	WARN	Charge current ≥ **160 A**	—	—	Charging only	✅
16	Discharge Over Current Protection	PROT	Discharge current ≥ **200 A**	✓	✓	Discharging only	✅
17	Discharge Over Current Warning	WARN	Discharge current ≥ **160 A**	—	—	Discharging only	✅
18	Charge Over Temperature Protection	PROT	Any cell temperature ≥ **50 °C**	✓	✓	Charging only	✅

ID	Alarm Name	Type	Trigger Condition (Default)	NOT	DECDLY	Active During	V1012
19	Charge Over Temperature Warning	WARN	Any cell temperature ≥ **45 °C**	—	—	Charging only	✅
20	Discharge Over Temperature Protection	PROT	Any cell temperature ≥ **50 °C**	✓	✓	Discharging only	✅
21	Discharge Over Temperature Warning	WARN	Any cell temperature ≥ **45 °C**	—	—	Discharging only	✅
22	Charge Under Temperature Protection	PROT	Any cell temperature ≤ **5 °C**	✓	✓	Charging only	✅
23	Charge Under Temperature Warning	WARN	Any cell temperature ≤ **10 °C**	—	—	Charging only	✅
24	Discharge Under Temperature Protection	PROT	Any cell temperature ≤ **5 °C**	✓	✓	Discharging only	✅
25	Discharge Under Temperature Warning	WARN	Any cell temperature ≤ **10 °C**	—	—	Discharging only	✅
26	Low Capacity Protection	PROT	State of Charge (SOC) < **0 %**	✓	✓	Always	✅
27	Low Capacity Warning	WARN	State of Charge (SOC) < **5 %**	—	—	Always	✅
29	Cell Unbalance Protection	PROT	Voltage spread across cells ≥ **400 mV**	✓	✓	Always	✅
30	High Pressure Warning	WARN	Ambient Pressure ≥ **30 kPa**	—	—	Always	✅
31	LIN Sensor Fault	PROT	LIN Bus errors ≥ **5** Consecutive *(see §4.1)*	✓	✓	Always	✅
32	Current Sensor Fault	WARN	Current sensor reads an invalid reading *(see §4.2)*	—	—	Always	✅
33	Temperature Sensor Fault	PROT	Temperature sensor reads an invalid reading *(see §4.2)*	✓	✓	Always	✅

ID	Alarm Name	Type	Trigger Condition (Default)	NOT	DECDLY	Active During	V1012
34	High H2 Protection	PROT	H_2 gas level from LIN sensor ≥ **6** (0–10 sensor scale, see §4.3)	✓	✓	Always	✅
35	High CO Protection	PROT	CO gas level from LIN sensor ≥ **6** (0–10 sensor scale, see §4.3)	✓	✓	Always	✅
47	Control Board Over Temperature Protection	PROT	Control board temperature ≥ **60 °C**	✓	✓	Always	✅
49	Delta Temperature Warning	WARN	Cell temperature abs(Max - Min) ≥ **5 °C**	—	—	Always	✅
52	Current Sensor Reverse Connection Warning	WARN	When the voltage direction is opposite to the current direction *(see §4.1)*	—	—	Always	✅
53	Low Piezo Warning	WARN	Any cell pressure sensor reading ≤ **30 kPa**	—	—	Always	✅
57	LOG Write Warning	WARN	SD card not detected	—	—	Always	✅
59	High Supply Warning	WARN	BMU supply voltage > **30 V**	—	—	Always	✅
60	Low Supply Warning	WARN	BMU supply voltage < **10 V**	—	—	Always	✅
61	Short Circuit Protection	PROT	Current magnitude ≥ **300 A**	✓	✓	Always	✅
62	Balance Break Protection	PROT	Balance resistor hardware state does not match software command	✓	✓	Always	✅

Unchecked errors in the software

ID	Alarm Name	V1012
1	Emergency Stop	❌
2	Over Voltage Protection	❌
3	Over Voltage Warning	❌

ID	Alarm Name	V1012
8	Under Voltage Protection	❌
9	Under Voltage Warning	❌
28	Low SOH Warning	❌
36	Pack Over Temperature Protection	❌
37	Pack Over Temperature Warning	❌
38	PBUSB Over Temperature Protection	❌
39	PBUSB Over Temperature Warning	❌
40	NBUSB Over Temperature Protection	❌
41	NBUSB Over Temperature Warning	❌
42	PMSD Over Temperature Protection	❌
43	PMSD Over Temperature Warning	❌
44	NMSD Over Temperature Protection	❌
45	NMSD Over Temperature Warning	❌
46	CPU Over Temperature Protection	❌
48	RB Over Temperature Protection	❌
50	Pressure Sensor Warning	❌
51	Piezo Sensor Warning	❌
54	CANBUS COM Warning	❌
55	USB COM Warning	❌
56	EEPROM Write Warning	❌
58	Insulation Resistance Warning	❌

Active in V1012 (39 alarms): 0, 4–7, 10–27, 29–35, 47, 49, 52–53, 57, 59–62
Not evaluated in V1012 (24 alarms): 1–3, 8–9, 28, 36–46, 48, 50–51, 54–56, 58

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4. Non-Standard Alarm Logic

Section 2 describes the standard alarm set: **PROT/WARN** holds a physical limit and **TH** holds a same-unit hysteresis margin. A handful of alarm sets repurpose one or both of these fields for different logic. This section documents those exceptions so the values in section 3 are not misread as physical thresholds.

4.1 Counter-Based Alarms

For these alarms, the PROT/WARN field is not a physical limit — it is a **cycle or event count**. The physical comparison, if any, lives in the TH field instead.

ID	Alarm Name	PROT/WARN Field Meaning	TH Field Meaning	Trigger Logic
31	LIN Sensor Fault	Number of consecutive LIN bus communication errors required to trigger	Hysteresis subtracted from the error count (not mV or any physical unit)	Triggers once the consecutive LIN error count reaches the PROT value. Clears once the error count falls below PROT minus TH.
52	Current Sensor Reverse Connection Warning	Number of consecutive evaluation cycles the fault condition must repeat	The actual voltage-drop/rise threshold, in **mV**, checked on every cycle	While charging, pack voltage is expected to rise; if it instead falls by more than the TH value (in mV) on a given cycle, an internal step counter increments. While discharging, the same logic applies in reverse (voltage expected to fall; a rise beyond TH increments the counter). Once the counter reaches the configured PROT number of consecutive cycles, the alarm triggers. Any cycle where voltage moves in the expected direction resets the counter to 0 and clears the alarm.

> **Reading the reference table:** for these two alarms, the value shown in the section 3 "Trigger Condition" column is a **count**, not a voltage or current value. The physical threshold that actually drives the comparison is the alarm's TH parameter.

4.2 Sensor-Validity Alarms (Enable Flag Only)

ID	Alarm Name	PROT Field Meaning	TH Field Meaning	Trigger Logic
32	Current Sensor Fault	Enable flag only; any value above the parameter minimum enables the check	Not used	Triggers when the current reading equals the sensor's invalid-value code while the alarm is enabled. There is no numeric comparison against the PROT value itself. When active, all current alarms (14–17) are forced inactive.
33	Temperature Sensor Fault	Enable flag only; any value above the parameter minimum enables the check	Not used	Triggers when any cell temperature reading equals the sensor's error code, or the battery-temperature-invalid flag is set, while the alarm is enabled. When active, all cell temperature alarms (18–25) are forced inactive.

For these two alarms, PROT does not gate a limit **—** it only turns the fault check on or off. Setting the parameter to its minimum value disables the check entirely, same as the disable convention in section 2.1, but there is no "limit" being disabled, only the validity check itself.

4.3 Gas Concentration Alarms (Sensor-Scale Units)

ID	Alarm Name	PROT Field Meaning	TH Field Meaning
34	High H2 Protection	Gas level reported by the LIN sensor, on its native **0–10 scale** (not ppm or mg/m^3)	Hysteresis margin on the same 0–10 scale
35	High CO Protection	Gas level reported by the LIN sensor, on its native **0–10 scale**	Hysteresis margin on the same 0–10 scale

These two follow the standard threshold/hysteresis pattern from section 2 **—** the only exception is the unit: the LIN gas sensor reports a **level from 0 to 10**, not a physical concentration. A PROT value of 6 means "level 6 on the sensor's own scale," not 6 ppm or any other physical unit.

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